stuff that falls into Gargantua long after Cooper falls in (long after, as measured by the external universe's time; Earth's time). If Cooper is hit by that singularity and survives, the universe's far future will be in his past. He will be so far in *our* future that, even with the help of bulk beings, he won't be able to return to the solar system until billions of years after he left, if ever. That would prevent him from ever reuniting with his daughter, Murph.

So Chris firmly chose Cooper to be hit by the outflying singularity, not the infalling one—hit by the singularity arising from stuff that fell into Gargantua *before* the Ranger, not after it.

Chris's choice, though, presents a bit of a problem for my scientist's interpretation of the movie. But not a problem so severe as backward time travel. If the Ranger falls directly into Gargantua from the critical orbit, then its infall is slow enough that the infalling singularity will catch up to it and hit it. For the Ranger to hit the outflying singularity instead, as Chris wants, the Ranger must nearly outrun the infalling singularity, which is descending at the speed of light. The Ranger can do so, if it is given a large, inward kick. How? The usual: by a slingshot around a suitable intermediate-mass black hole soon after leaving the *Endurance*.

What Does Cooper See Inside Gargantua?



Looking up as he falls inward, Cooper sees the external universe. Because his infall has been sped up, he sees time in the external universe flow at roughly the same rate as his own time,[51] and he sees the image of the external universe reduced in size,[52] from about half of the sky to roughly a quarter.